

	31	A
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	34	C
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	35	D
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Question	Marking guidance	Additional comments/guidelines	Mark
05.1	0.86 (Wb s ⁻¹) ✓	Allow 0.85 or 0.87	1
05.2	Induced emf caused by change in flux linkage ✓ Zero emf (between 1.90 and 1.95 s) because no change in flux linkage OR equal magnitudes of emf because (trolley at) constant speed ✓ And one from: ✓ Explains emf increases or decrease in terms of <u>rate of</u> change of flux linkage Explains change in sign in terms of Lenz's Law	Condone "flux" for "flux linkage" Condone reference to 'peak(s)'. If no mark given, allow 1 mark for idea that induced emf is caused by a change in magnetic field with coil	3
05.3	Uses $v = s/t$ with either 1.5, 2.5, 4.0 or 5.5 cm or with a duration taken from the graph ✓ 0.5 (m s⁻¹) ✓		2
05.4	Larger emf (for either peak) ✓ Narrower waveform (for either peak) ✓ Any two from: ✓ ✓ 2nd peak higher than 1st 2nd peak narrower than 1st Shorter separation between peaks	Sketch must start at 1.42 s for full marks	4
Total			10

Question	Key	Question	Key
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	31	B
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	34	B
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	35	C
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	use of at least two independent relevant equations, one including Φ ✓ $\text{kg m}^2 \text{s}^{-2} \text{A}^{-1}$ ✓	Expect to see eg $N\Phi = NBA$ and $F = BIL$ and $F = ma$ An alternative route could be $\varepsilon = \frac{\Delta N\Phi}{t}$ and $V = \frac{W}{Q}$ and $W = \frac{1}{2}mv^2$ full marks for correct final answer, units can be in any order ignore 'turns' in final answer	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	$(N)\Phi = (N)BA$ and $\varepsilon = \frac{\Delta(N)\Phi}{\Delta t}$ both seen OR gradient = $\frac{\Delta B}{\Delta t}$ ✓ all combined ✓	reject $\frac{B}{t}$ and $\frac{(N)\Phi}{t}$ without Δ or d symbols	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Attempt to determine gradient (of a tangent) and substitutes their gradient into equation ✓ max gradient found with evidence seen on graph ✓ 0.21 to 0.23 (V) ✓	For MP2, evidence must be a tangent drawn larger than half the graph's length/height; values taken from tangent must be more than half the axes' length apart expect to see gradient values in the range of 2.0 to 2.2	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	(Very) large emf induced (at first and then drops to zero) ✓ opposite sign to the first reading ✓ emf related to large rate of change of flux linkage OR opposite sign related to Lenz's law ✓	allow reading has minus value condone rate of change of flux density in coil X	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	laminations reduce energy loss/increase efficiency ✓ laminations limit/reduce (the magnitude of) the eddy currents (in the iron core)✓	condone reduce heating effect for MP1 do not allow 'prevents' or 'eliminates' eddy currents	2	AO4
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
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7	C
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19	A
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	There is an increase/change in the magnetic flux linkage with the coil.✓ Reference to Faraday's Law.✓	Do not accept 'increase in rate of change of flux linkage'	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	(Ref to Lenz's Law) Emf is induced in direction to oppose changing flux linkage as magnet enters/leaves coil.✓ Flux linkage increasing and decreasing are opposite effects (so emfs must have opposite signs) ✓	Condone 'flux' for 'flux linkage'	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Magnet is going slower when entering ✓ Smaller rate of change of flux linkage ✓ (Therefore smaller induced emf) (Reference to Faraday's Law)	Argument can be made for V_1 smaller or V_2 greater.	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Magnet is accelerating downwards ✓ Time from top of coil to centre of coil is greater than time from centre to bottom of coil. ✓ Emf is zero when magnet is at the centre of the coil. ✓		Max 2	AO4
Total			8	

Question	Key	Answer	AO
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11	B	$\frac{N\Phi}{t_2}$	AO1
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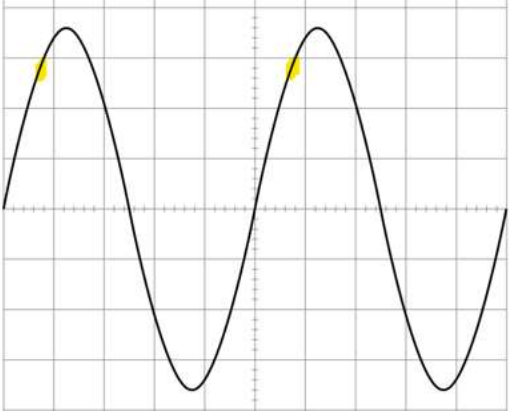
21	C	2 and 3
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19	B	150 W	AO2
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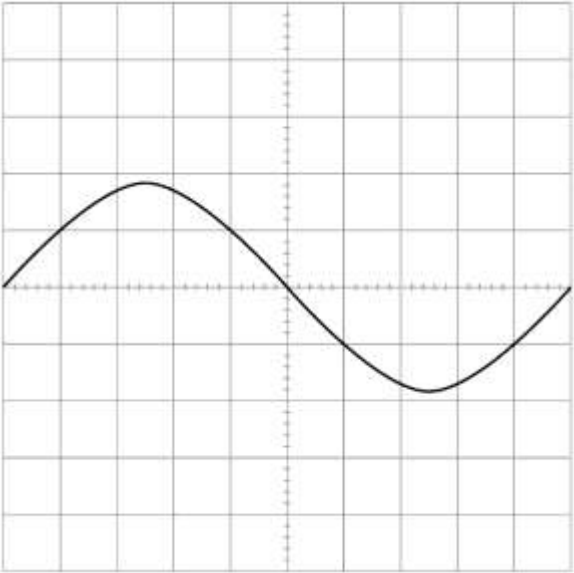
20	C	2.0 A	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	(Use of flux linkage = $BAN \cos \theta$) 121 (turns) ✓	Accept 2sf 120 turns. Must be whole number.	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Idea that as angle changes flux linkage changes ✓ States and applies Faraday's law eg an emf is induced when the flux (linkage) changes ✓	Accept perpendicular area or effective area for angle. Allow alternative MP2 eg 'emf is proportional to rate of change of flux linkage' Do not award MP2 for an equation without support. Allow equation for Faraday's law provided N is included or Φ is defined as the 'flux linkage'.	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	X positioned on the trace 0.7 of a square from start or equivalent ✓ Evidence of calculation eg Position on time axis is 5 squares $\times 51 \div 360$ OR position on V axis = $V_{\max} \times \sin(51^\circ)$ ✓	Allow X on the line above the middle of the square by eye 	2	AO2 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Determines period or number of squares for one period ✓ Determines peak emf or number of squares for peak ✓ Time base = 4 (ms cm⁻¹) y-gain = 0.5 (V cm⁻¹)	Expect to see 20 (ms) or 5 squares Expect to see 1.8 (V) or 3.6 squares Only accept 1 sf answers	4	AO3 $\times$ 2 AO4 $\times$ 2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.5	Sinusoidal with any initial phase AND half the height of Figure 14 ✓ double the period of Figure 14 ✓	 For multiple curves judge the first cycle for how sinusoidal it is. Ignore vertical shift in MP2.	2	AO3
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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20	C	Their size depends on the resistivity of the material of the core.	AO1

18	C	
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19	A	
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20	D	
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21	c	
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22	c	3200 V	AO2
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23	D	5.7 A	AO1
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question	marking guidance	mark	comments
05	The marking scheme for this question includes an overall assessment for the quality of written communication (QWC). There are no discrete marks for the assessment of QWC but the student's QWC in this answer will be one of the criteria used to assign a level and award the marks for this question. Descriptor – an answer will be expected to meet most of the criteria in the level descriptor. Level 3 – good -claims supported by an appropriate range of evidence -good use of information or ideas about physics, going beyond those given in the question -argument well-structured with minimal repetition or irrelevant points -accurate and clear expression of ideas with only minor errors of grammar, punctuation and spelling Level 2 – modest -claims partly supported by evidence, -good use of information or ideas about physics given in the question but limited beyond this - the argument shows some attempt at structure -the ideas are expressed with reasonable clarity but with a few errors of grammar, punctuation and spelling Level 1 – limited -valid points but not clearly linked to an argument structure -limited use of information about physics -unstructured -errors in spelling, punctuation and grammar or lack of fluency Level 0 -incorrect, inappropriate or no response	6	Selects two from • Eddy currents • Hysteresis • Copper losses • Flux leakage For each one, student should explain how the losses occur how losses are minimised by design and choice of materials.

question	marking guidance	mark	comments
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